

CGMega explainable graph neural network framework with attention mechanisms for cancer gene module dissection

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Research Interests

Biomedical Image Understanding,
Biological Language Modeling,
Machine Learning and Data Mining

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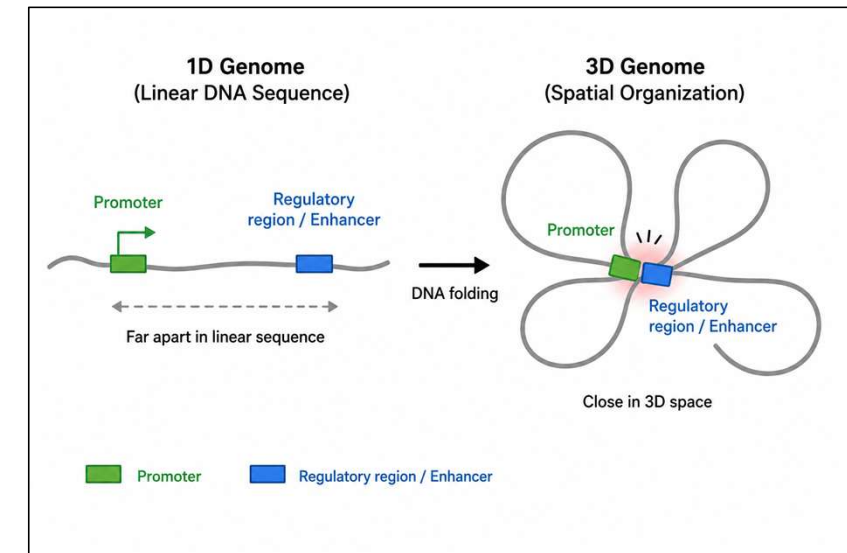
Problem

- How to encode Hi-C data and integrate with other multi-omics data challenge the precise detection of gene modules.
- Deciphering the high-order relationships that are embedded in gene modules remains challenging.

What is Hi-C?

Hi-C is a technology used to analyze the 3D structure of the genome by measuring how closely genomic regions are positioned within the nucleus.

(using for cancer research & gene expression regulation studies.)



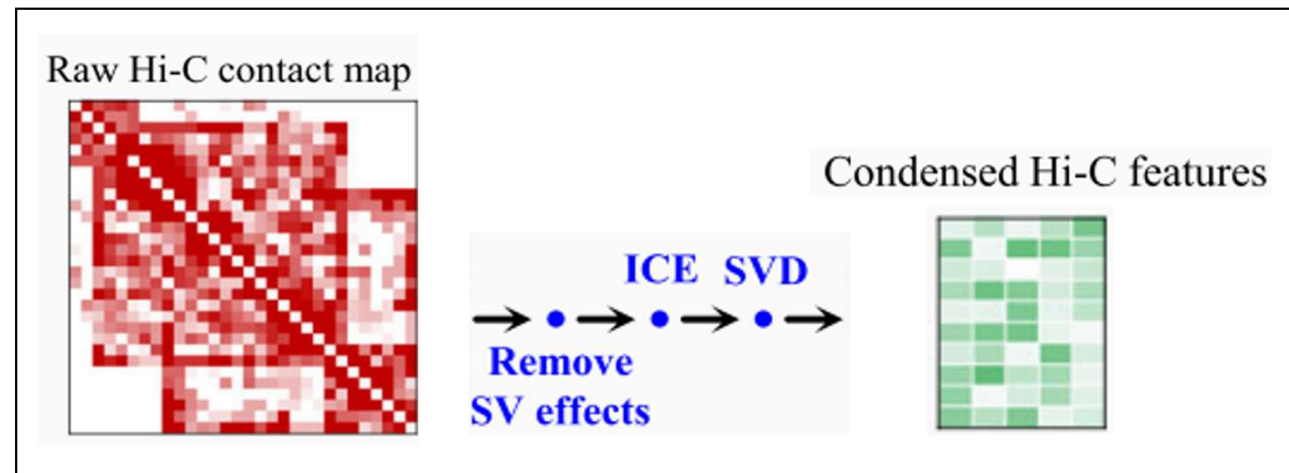
Hi-C Data Collection Methods

Normalization

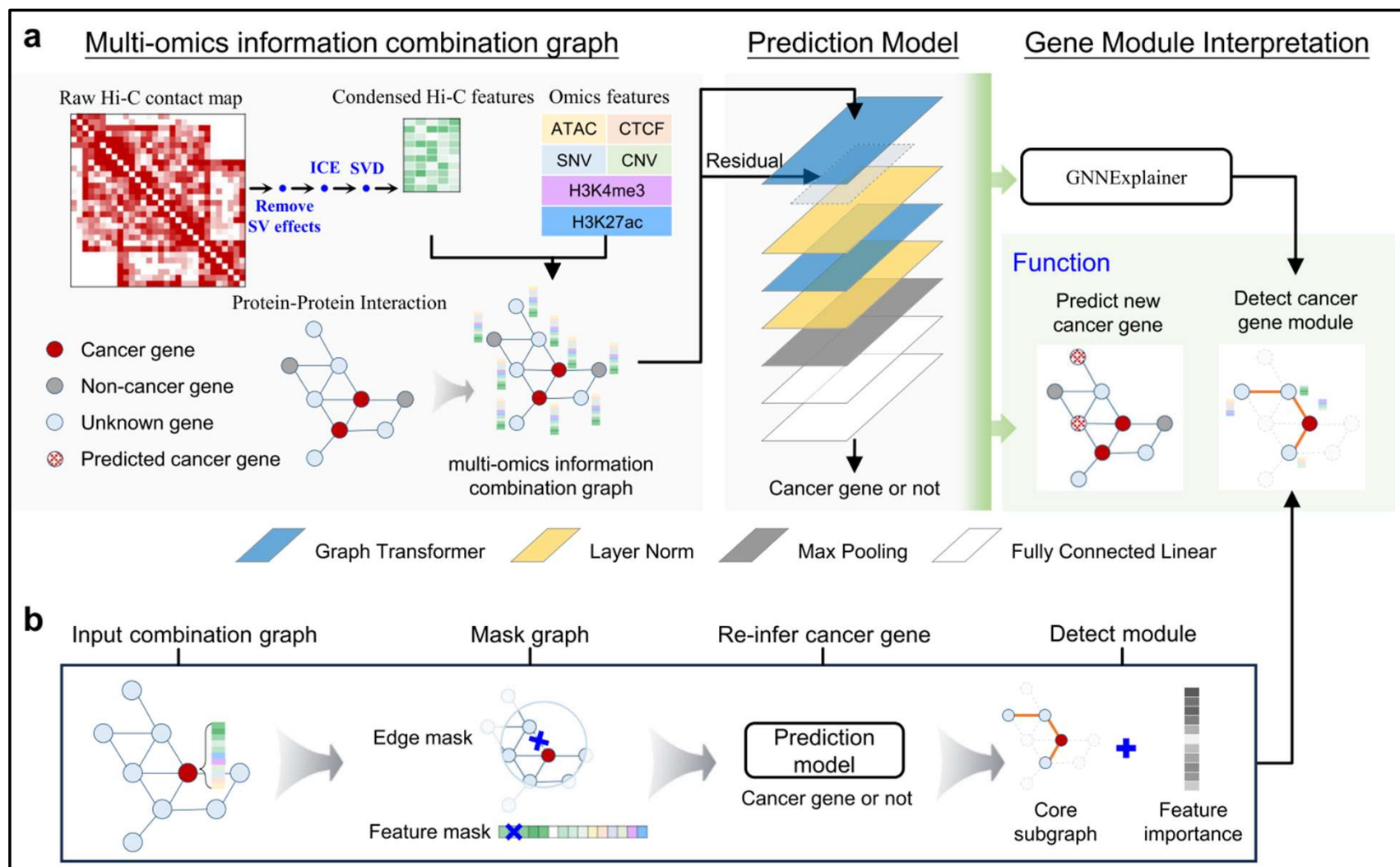
iterative correction and eigenvector decomposition(ICE)

Singular vector decomposition(SVD)

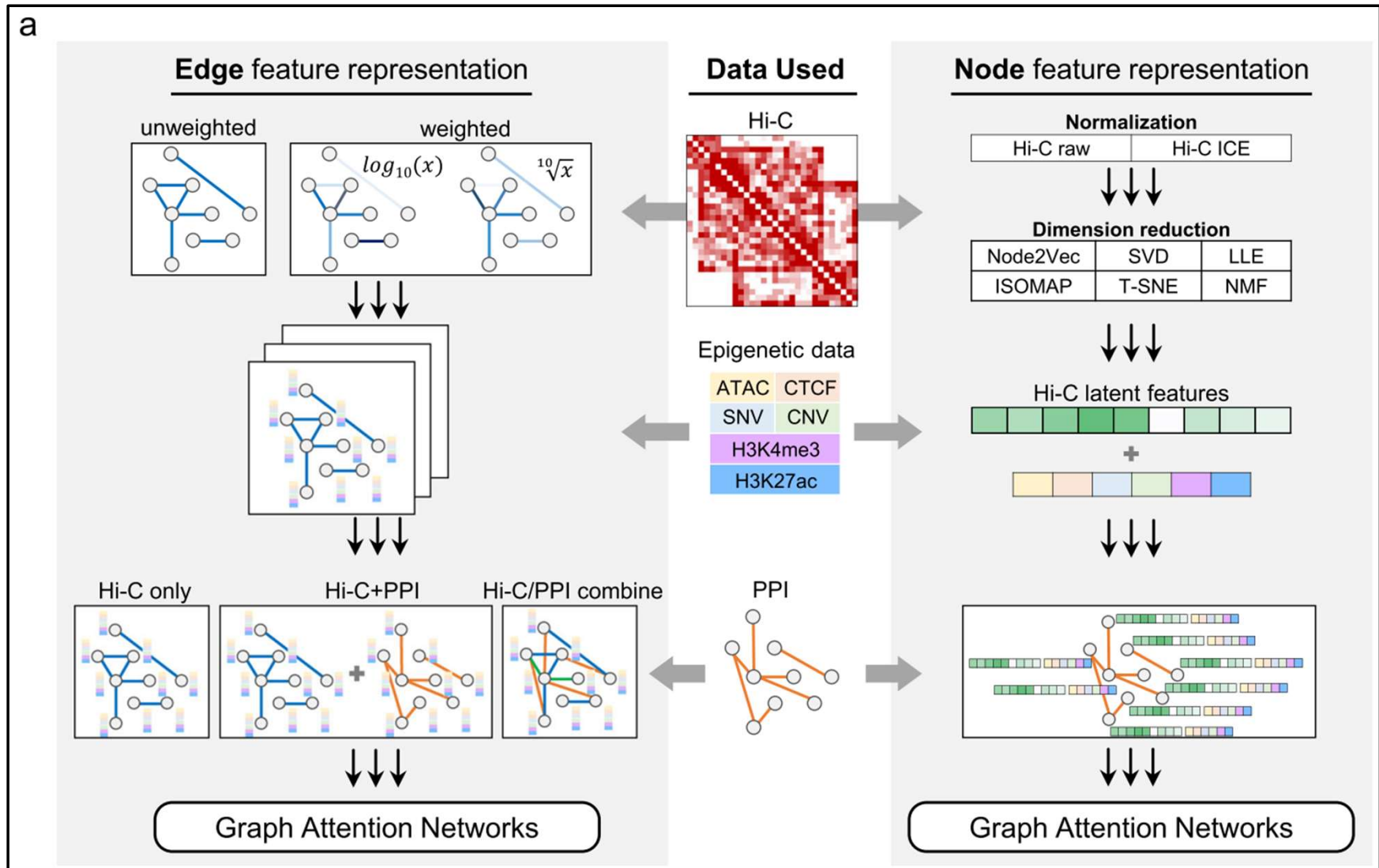
Decompose normalized Hi-C data into five dimensions as gene features



Overview of CGMega framework



New strategy for Multi-omics data integration



Performance evaluation of multi-omics data integration approaches

b Edge feature representation

		AUPRC
Hi-C only	unweighted	0.4649
	weighted-log ₁₀ (x)	0.4944
	weighted-square	0.4691
Hi-C+PPI	unweighted	0.8276
	weighted-log ₁₀ (x)	0.8811
	weighted-square	0.8789
Hi-C/PPI combine	unweighted	0.9004
	weighted-log ₁₀ (x)	0.8908
	weighted-square	0.9028
PPI only		0.8964

c Node feature representation

	AUPRC
Node2Vec	0.9114
SVD	0.9140
LLE	0.8899
ISOMAP	0.8890
NMF	0.9087
t-SNE (1)	0.8984
t-SNE (2)	0.8959
t-SNE (3)	0.9061

d Node feature representation with SVD

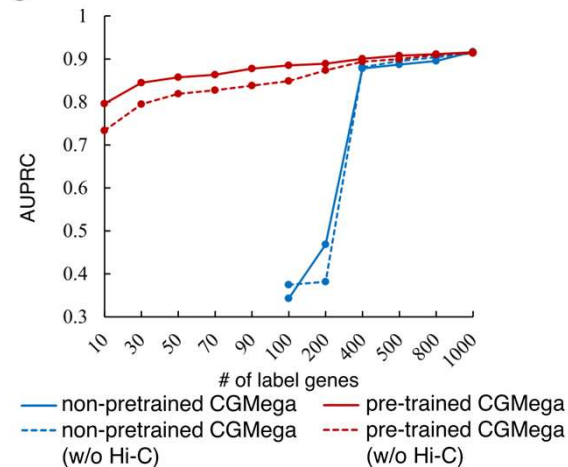
	AUPRC
ICE-1	0.9020
ICE-2	0.9109
ICE-5	0.9140
ICE-10	0.9133
RAW-1	0.8965
RAW-2	0.9115
RAW-5	0.9029
RAW-10	0.9042

CGMega performance in cancer gene prediction task

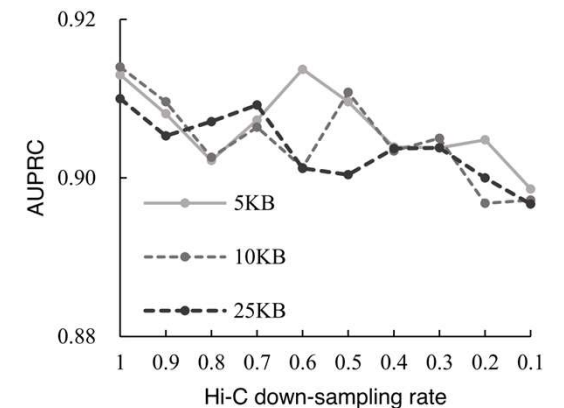
b

		AUPRC	AUROC	ACC	F1
Ours	CGMega	0.9140	0.9630	0.9216	0.8081
	MTGCN	0.8290	0.9417	0.9052	0.7604
	EMOGI	0.7419	0.8994	0.6784	0.5215
GNNs	MODIG	0.6326	0.5001	0.5182	0.1957
	GCN	0.6876	0.8914	0.8557	0.6429
	GAT	0.7752	0.9355	0.8845	0.7308
	MLP	0.8501	0.9331	0.9155	0.7515
N2V	SVM	0.8731	0.9408	0.9155	0.7657
	MLP (w/o PPI)	0.4006	0.7658	0.8165	0.2393
	SVM (w/o PPI)	0.4000	0.6704	0.8247	0.1584

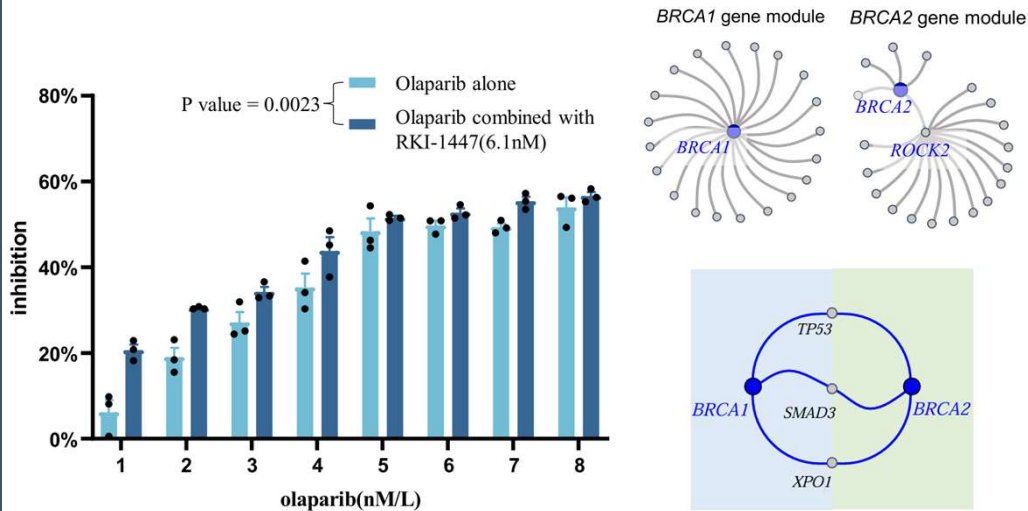
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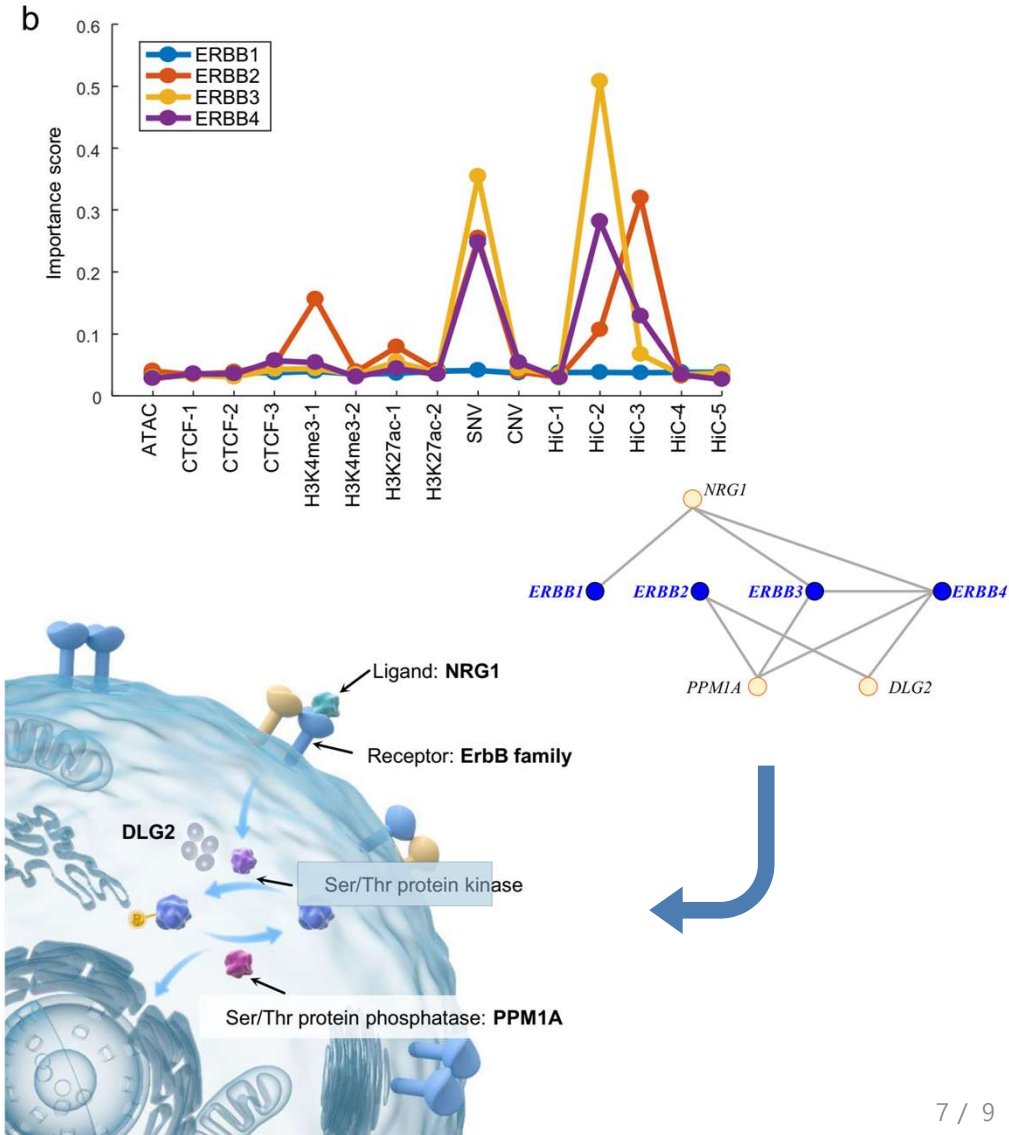
e



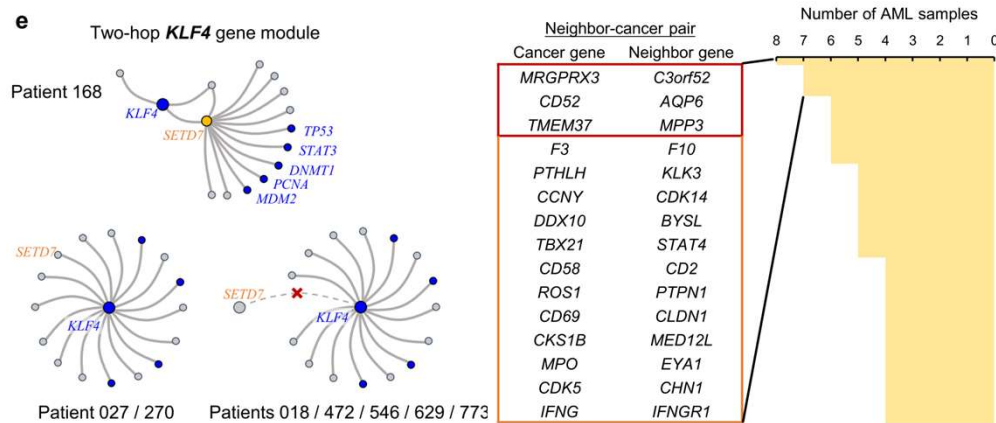
Gene modules in breast cancer cell line



Gene modules of ErbB family



Gene modules in AML patients



Take-home Messages

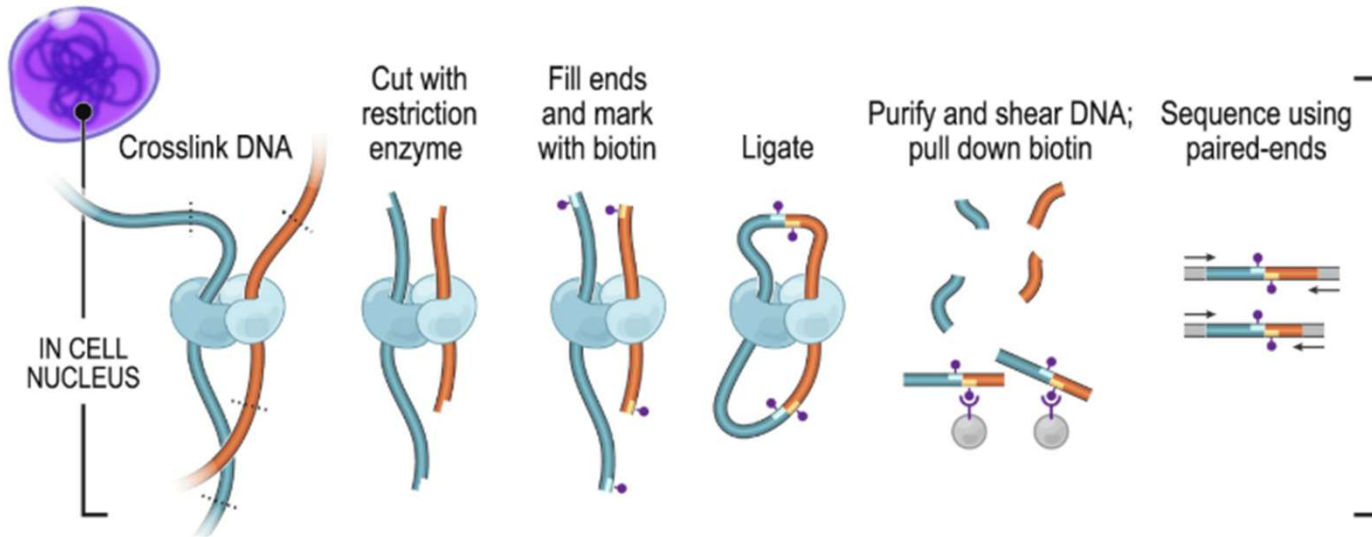
- Encoding 3D genome structural information (Hi-C data) as node features rather than gene connectivity structures improved the model's predictive performance.
- Incorporating Hi-C data also contributed to few-shot learning, enabling effective training with fewer than 200 known cancer genes.
- By dissecting gene modules, the model identified which features contributed most to cancer gene prediction. Assigning different contribution weights to each gene module enabled more precise interpretation of cancer gene causality.

Conclusion

- This study demonstrated various techniques for interpreting model predictions and showed how highly accurate and precise predictions can be achieved even with limited training data.

Thank you

Appendix: Hi-C process



Hi-C reveals 3D genome structure

TADs (Topologically Associating Domains):

regions with strong internal interactions and weak external interactions

Chromatin loops:

physical contacts between distant regulatory elements (e.g., enhancer–promoter)

A/B compartments:

A = active chromatin, B = inactive chromatin

Importance in bioinformatics and cancer research

- Links enhancer mutations to promoter effects
- Reveals 3D genome reorganization in cancer
- Supports gene regulatory network inference

Appendix: ICE

기본 원리: 모든 genomic bin(matrix 형태)은 평균적으로 같은 총 contact 수를 가져야 한다

Iterative correction scaling을 반복 수행: (논문에서는 여기까지만 한 후 SVD로 넘어감)

1. contact matrix 초기값 사용
1. 각 row/column의 총합 계산
2. 각 bin을 normalization factor로 나눔
3. 다시 row/column 재계산
4. 수렴할 때까지 반복

결과: 모든 bin의 "bias 제거된 균형 상태" 도달

Eigenvector decomposition (A/B compartment): ICE로 정규화된 matrix에 대해 eigenvector decomposition 수행 (논문에서는 이 과정을 거치진 않음)

- 입력: Hi-C contact matrix (bin × bin)
- 출력: eigenvector (각 bin에 대한 값)

Appendix: Hi-C Data Collection Methods

Multi-omics feature calculation. First, CGMega transfer original multi-omics data to input embedding according to Data Collection and Preprocessing. In detail, ATAC-seq, H3K4me3, H3K27ac, and CTCF ChIP-seq data are processed by the computeMatrix method of deepTools⁸⁸. CNV and SNV frequency are calculated as the following equation:

$$\mathbf{x}_i^{CNV} = \frac{\sum_{s \in S} \mathbf{v}_{s,i}^{CNV}}{|\mathbf{S}|}, \quad (2)$$

$$\mathbf{x}_i^{SNV} = \frac{\sum_{s \in S} \mathbf{v}_{s,i}^{SNV}}{|\mathbf{S}| * \text{len}(\text{Gene}_i)}, \quad (3)$$

and $\text{len}(\text{Gene}_i)$ denotes the length of gene i . The raw Hi-C matrix is corrected by NeoLoopFinder and ICE, and then a gene contact matrix C is derived by mapping bin locations to genes:

$$\mathbf{C} = \begin{bmatrix} c_{11} & \cdots & c_{1n} \\ \vdots & \ddots & \vdots \\ c_{n1} & \cdots & c_{nn} \end{bmatrix}, \quad (4)$$

$$c_{ij} = \sum_{a=m'}^m \sum_{b=n'}^n h_{ab}, \quad (5)$$

where h_{ab} denotes elements in Hi-C matrix H , c_{ij} denotes elements in gene contact matrix C and gene i and j are located within the range $[m', m]$ and $[n', n]$ in the chromosome respectively. Finally, the gene contact matrix C is decomposed to 5 dimensions using SVD. All the features after normalization are concatenated together to form the input feature matrix $\mathbf{X}$ as follows:

$$\mathbf{x}_i = [\mathbf{x}_i^{ATAC}, \mathbf{x}_i^{CTCF}, \mathbf{x}_i^{H3K4me3}, \mathbf{x}_i^{H3K27ac}, \mathbf{x}_i^{CNV}, \mathbf{x}_i^{SNV}, \mathbf{x}_i^{HiC}], \quad (6)$$

$$\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n]^T. \quad (7)$$

Appendix: GAT-based cancer gene prediction model

$$\mathbf{x}'_i = \frac{\mathbf{x}_i - E(\mathbf{x}_i)}{\sqrt{\text{Var}(\mathbf{x}_i) + \epsilon}} \odot \boldsymbol{\gamma} + \boldsymbol{\beta},$$

$$\begin{aligned}\mathbf{x}'_i &= \text{GraphTransformerConv}(G, \mathbf{x}_i) \\ &= \mathbf{W}_1 \mathbf{x}_i + \sum_{j \in N(i)} \alpha_{ij} \mathbf{W}_2 \mathbf{x}_j,\end{aligned}$$

$$\alpha_{ij} = \frac{(\mathbf{W}_3 \mathbf{x}_i)^\top (\mathbf{W}_4 \mathbf{x}_j)}{\sqrt{d}},$$

$$\mathbf{x}_i^{(l+1)} = \text{GraphTransformerConv}(G, [\mathbf{x}'_i, \mathbf{x}_i]),$$

Graph transformer layers: perform GAT mechanism

Layer-norm layer: The output of each graph transformer layer is passed through a corresponding layer-norm layer to normalize the activation

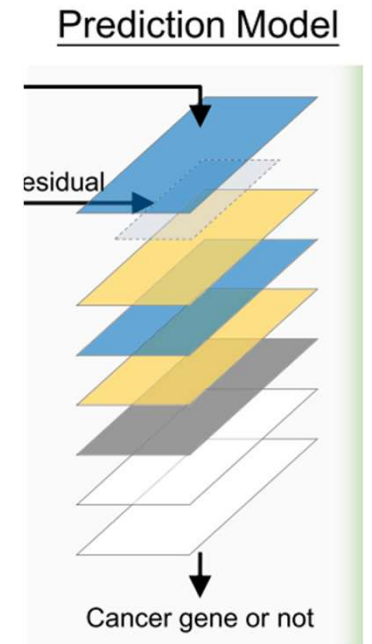
Max-pooling layer: To reduce the dimensionality of the features extracted by using the graph transformer layers.

Fully connected linear layer: To make the final prediction

1st: Transforming the data into a 32-dimensional representation

2nd: Producing the final output

Dropout layer: To prevent overfitting



Appendix: GNNExplainer

Framework

$$\max_{G_S} \mathbf{MI}(Y, (G_S, X_S)) = \mathbf{H}(Y) - \mathbf{H}(Y | G = G_S, X = X_S).$$

Subgraph, Feature

$$G_{Si} = A_i \odot 1\{M_i \geq \text{edge threshold}\}, \quad (12)$$

$$X_{Si} = \{x'_j | j \in G_{Si}\}, \quad (13)$$

where $x'_j = x_j \odot 1\{f_i \geq \text{feature threshold}\}$

Appendix: FigS1a, S1b,2b,2c,2e,2f,2g,2h

Figure S1

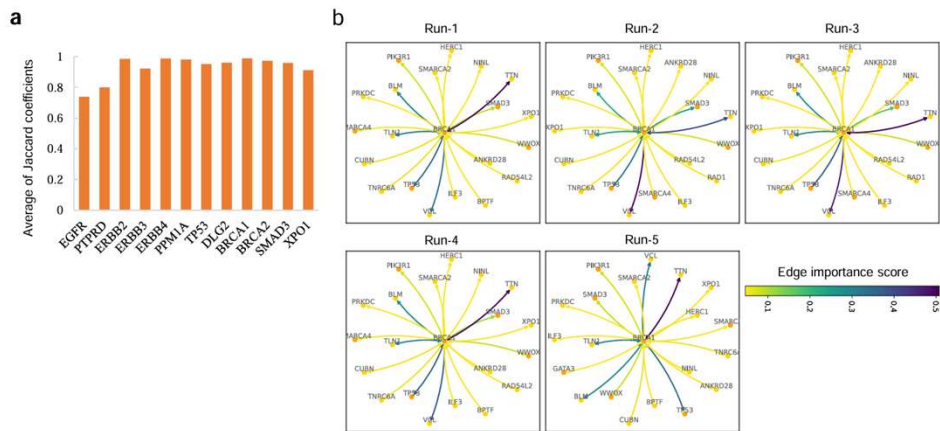
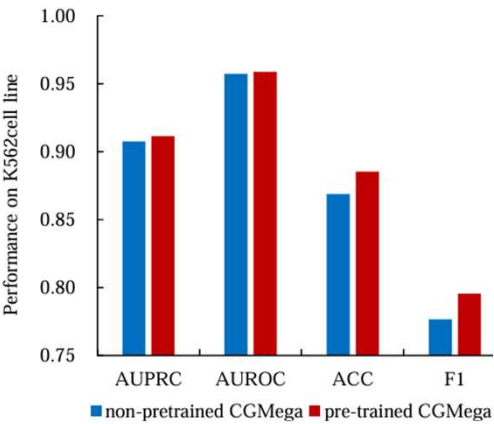


Fig. S1 robustness of GNNExplainer in CGMega. **a** We repeated GNNExplainer five times on 12 randomly selected cancer genes, and calculated the average Jaccard coefficients between any two repeats. **b** Taken BRCA1 gene module as an example to show the robustness of GNNExplainer interpretation. Source data are provided as a Source Data file.

Performance of Transfer learning (Supplementary Fig.S2b)



Performance of few-shot Transfer learning (Supplementary Fig.S2c) (y: AUPRC)

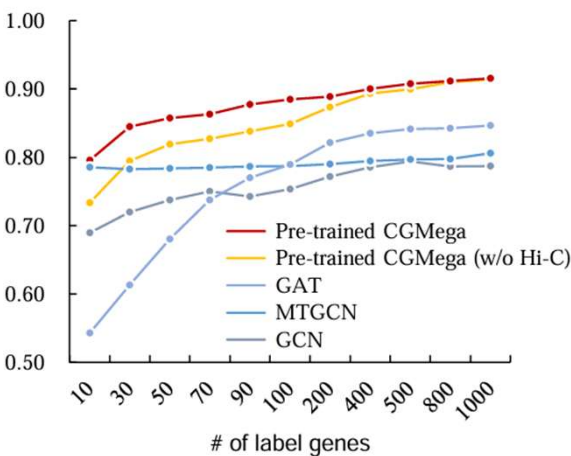
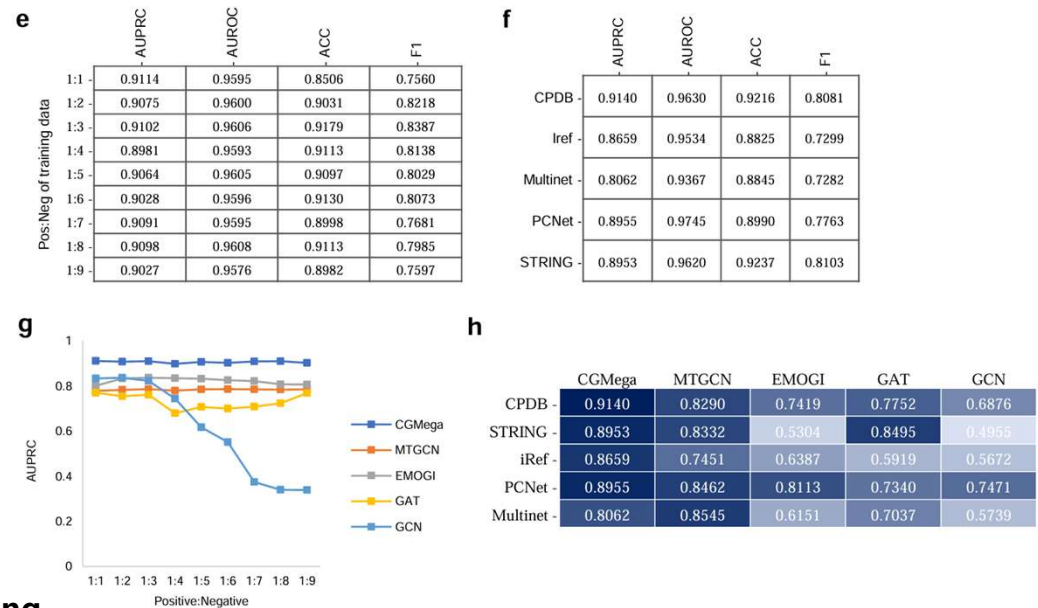
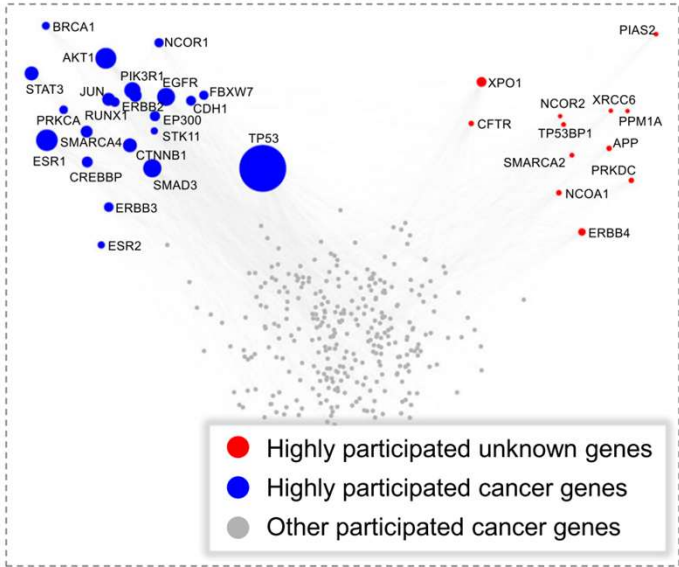


Fig.S2 e,f,g,h

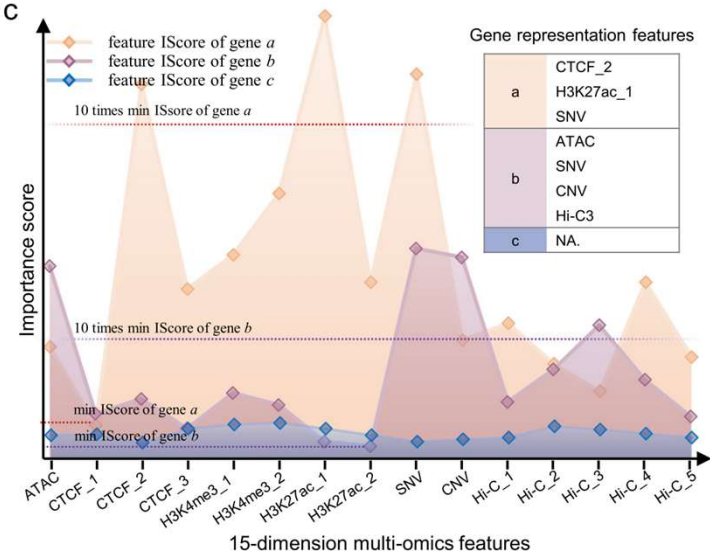


Appendix: Fig.4a,b,c,e,f,g

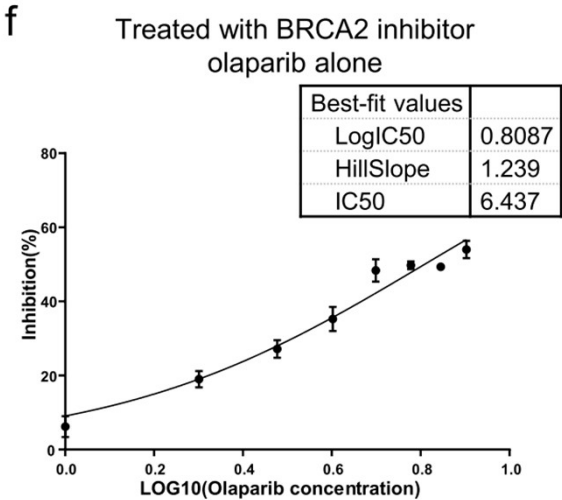
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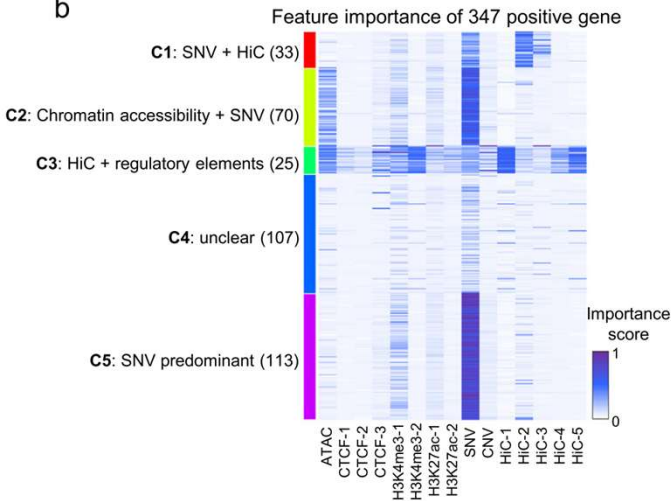
c



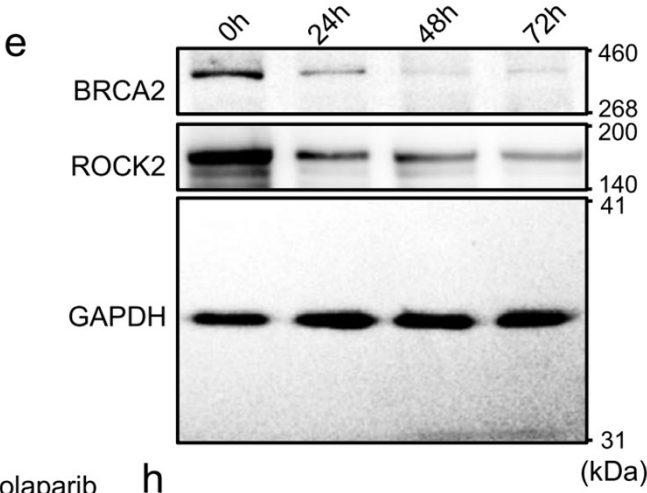
f



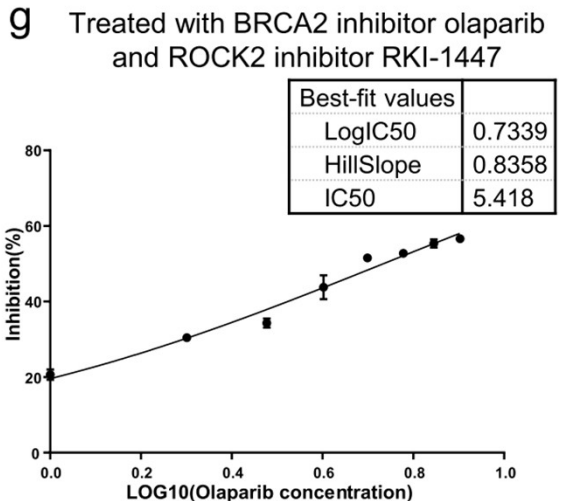
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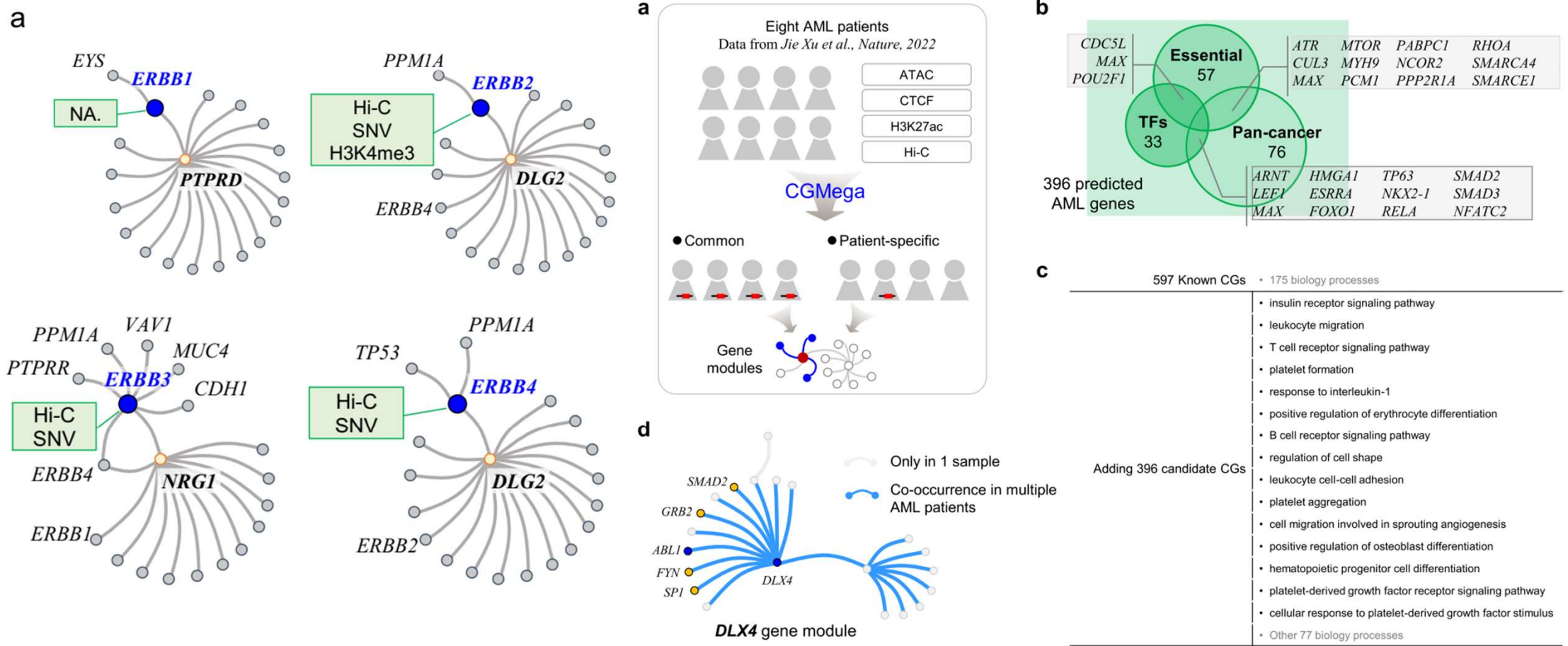
e



g

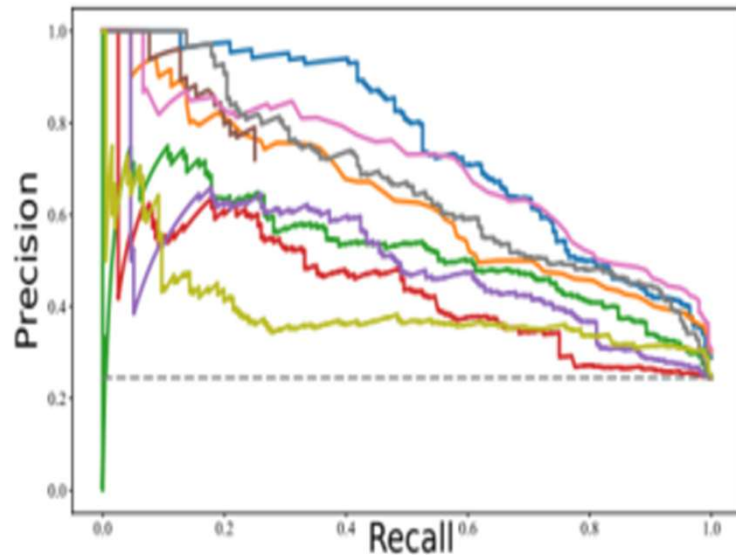


Appendix: Fig.5a, Fig.6a,b,c,d



Appendix

AUPRC (Area under the precision-recall curve)



$$\text{Precision} = \frac{TP}{TP + FP} \quad \text{Recall} = \frac{TP}{TP + FN}$$

TP: True positive

FP: False Positive

FN: False Negative

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

The **F1 score** is a metric that simultaneously reflects how thoroughly a model identifies relevant cases (Recall) and how accurately its predictions are made (Precision).